

Channel Packer User Guide

For Version 1.0.2

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Introduction

Channel Packing Theory

Modern game engines often use physically based rendering (PBR) maps in their pipeline to achieve more realistic material shaders. They often use greyscale (linear colour) images to represent different properties like metalness or specular.

However, having this many texture files impacts memory so they're often stored in a single RGB image as a single channel.

The standard layout for this in Unreal is occlusion, roughness and metallic (ORM) on the red, green and blue channels, respectively. However, many variations exist and the alpha channel can be used for height, masks, temporal offsets, etc.

More information on this can be found here:

<http://wiki.polycount.com/wiki/ChannelPacking>

The usual method

The usual method requires packing in either Photoshop or a Photoshop alternative by pasting into specific channels. Or by importing the maps into Substance Designer or Painter, which usually requires making your own template, and exporting from there. This is unnecessarily time-consuming and has been a personal annoyance for me.

My tool aims to streamline this process, along with unpacking.

Installation Guide

Itch.io (exe version)

1. Download the tool from the following:
<https://be-a-fish-studios.itch.io/spiders-channel-packing-tool>
2. Press 'Download now' and 'No thanks' to the donation
3. Unzip the folder and open the .exe file.
4. If you get any errors, ensure the Channel_Packer_Tool folder is in the same directory as the executable file, or it won't run.

Spiders Channel Packing Tool

This tool is a standalone channel packing tool used to merge separate RGB(A) images into a single image and unpack images into separate RGB(A) channels. This is a common texture optimisation technique. It was made to streamline PBR workflows.

This tool is FREE AND OPEN SOURCE! Developed as a uni project for my MSc. Only donate if you want to support me. You can use this for free on commercial projects. You do not need to credit me. (though that would be nice.) You can also modify it if you want. The source code and an alternative download are available on my GitHub.

https://github.com/be-a-fish/Spiders_Channel_Packer

No AI was used in the creation of this software. I do NOT give my consent for it to be used in AI training.

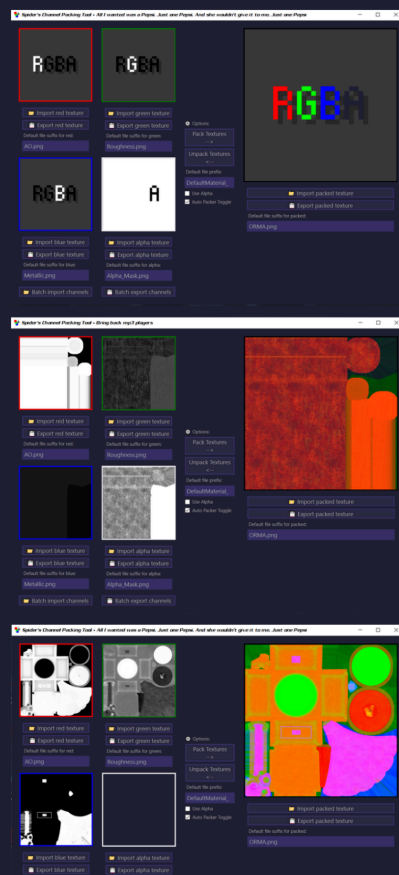
[More information](#) ▾

Download

[Download Now](#) Name your own price

Click download now to get access to the following files:

Spiders Channel Packer 1.0.zip 64 MB



Github (exe version or source code)

1. Go to https://github.com/be-a-fish/Spiders_Channel_Packer and look for the latest release on the right-hand side.
2. Download the zip for the exe or source code if you plan on opening it through Python.

The screenshot shows the GitHub repository page for 'Spiders_Channel_Packer' by user 'R-Luff-Escape-Studios'. The repository has 0 stars, 0 watchers, and 0 forks. The 'Releases' section shows a single release, 'release 1.0.0', marked as 'Latest' and released 5 hours ago. The 'Assets' section lists three files: 'Spiders.Channel.Packer.1.0.zip' (64.9 MB, 6 hours ago), 'Source code (zip)' (6 hours ago), and 'Source code (tar.gz)' (6 hours ago). The release description mentions adding a functional executable and future optimization goals.

☆ 0 stars

👁 0 watching

🔗 0 forks

Releases 1

📦 **release 1.0.0** Latest

5 hours ago

Packages

No packages published

[Publish your first package](#)

release 1.0.0 Latest

Compare ✎ 📦

R-Luff-Escape-Studios released this 5 hours ago

· 4 commits to main since this release exe ↻ bed472c

Added a functional executable.

I would like to optimise code a bit further in the future, as 64MB (while being my lucky number) is larger than I want for a lightweight program. I'm sure there's some unnecessary code in there too. That being said, I need to allocate my time accordingly and need to prioritise the user guide, which will be in the next release

▼ **Assets** 3

📦 Spiders.Channel.Packer.1.0.zip	64.9 MB	6 hours ago
📄 Source code (zip)		6 hours ago
📄 Source code (tar.gz)		6 hours ago

😊

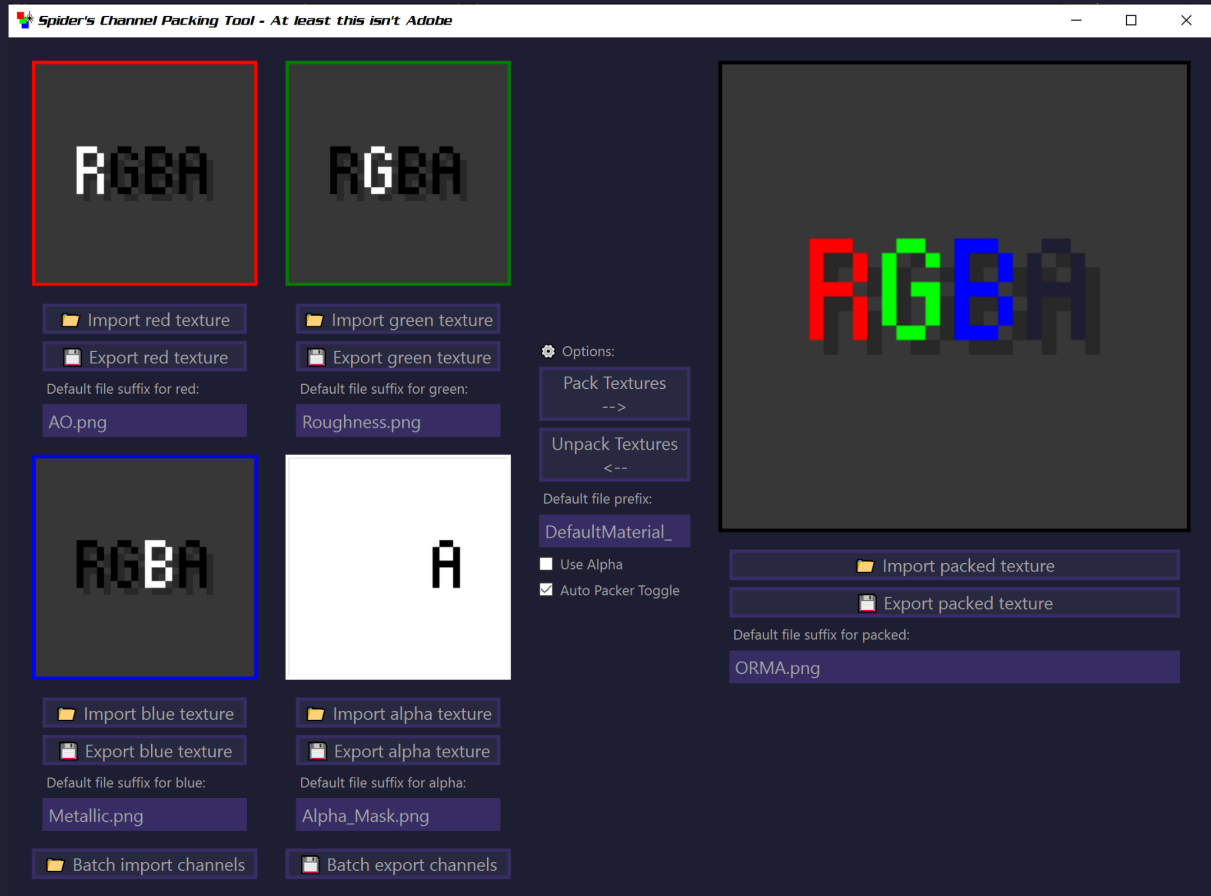
3. If opening the Python version, you may need to install some libraries. A full list is available in the credit section at the end of this guide with their install commands. To install them open terminal or command prompt and type the following command for all libraries:

`pip install library_name` – see page 17

UI Layout

Basic layout

The tool is split into 3 sections: the RGBA channels to the left, the packed image to the right, and the options section in the middle.



The batch import and export options are located under the separated RGBA channels

Image channel layout

Each image has the same layout, so the user only has to learn it once. This includes the packed image.

Default file name suffix for selected channel: Added to the prefix for batch importing/exporting



Import red texture

Export red texture

Default file suffix for red:

AO.png

Image texture:
The texture used for the channel. Outlined in its corresponding colour

Import button:
Opens import window

Export button:
Opens export window

Options layout

The options menu contains the basic user options for the tool which apply to all channels. Default file Prefix: added to the suffix in batch import/export options

Use Alpha toggle: when disabled, alpha channel is not packed and image is exported in RGB colour mode

Options:

Pack Textures

-->

Unpack Textures

<--

Default file prefix:

DefaultMaterial_

☐ Use Alpha

☒ Auto Packer Toggle

Pack texture button: Packs RGB(A) channels into the packed texture slot

Unpack texture button: Unpacks the packed texture into the RGB(A) texture slots

Auto Packer Toggle: When enabled automatically packs and unpacks textures on import

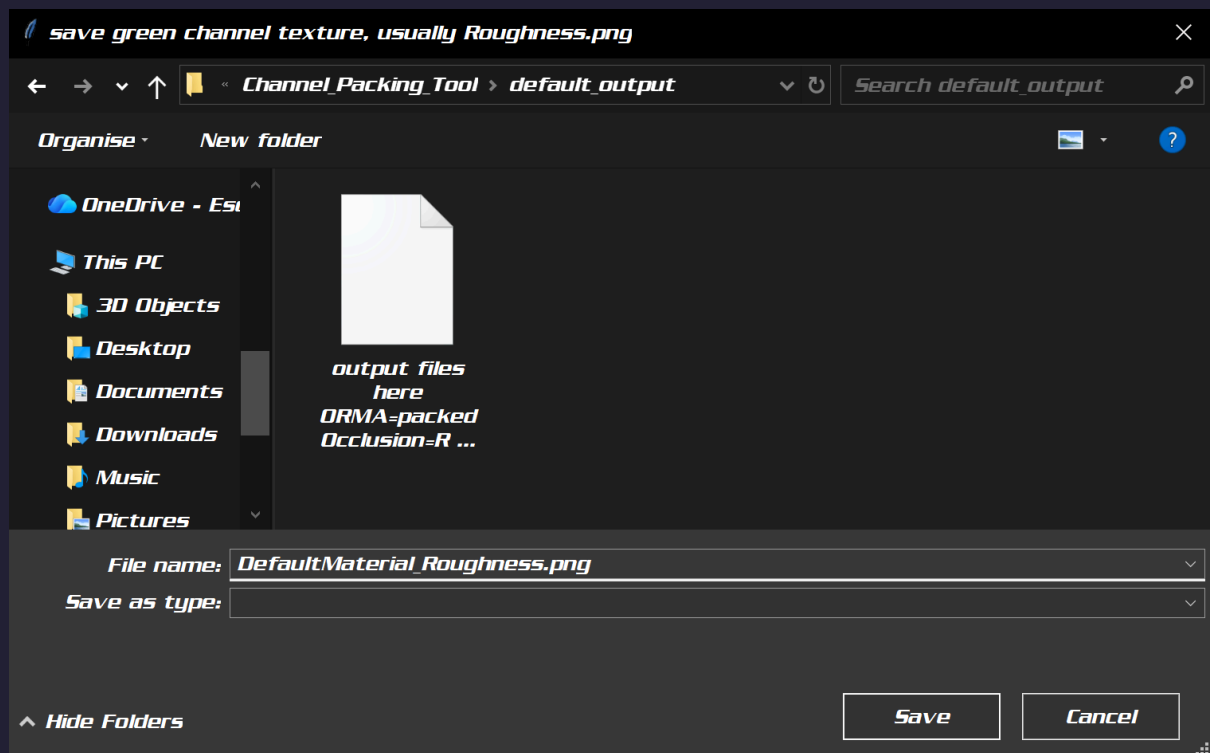
Export window

The save window should match most other software.

The file-type box is currently non-functional, so the user needs to type out their desired extension, but it defaults to .png if left blank.

See also Default path in troubleshooting if you press cancel or X

Window title gives a hint on the channel being saved



Default names – for advanced users

Each filename will be saved with a user-defined name on export; however, advanced users may need to export multiple files with a set format for their pipeline. For example, T_WeaponName_Roughness.png.

This is not fully standardised, but most pipelines use an ORM or ORMA (Occlusion, Roughness, Metallic, Alpha) layout, so I've done my best to include relevant default file extensions. A user should choose ones which relate to their project.

The file name is calculated as follows: (Prefix)+(Suffix). Please note the file extension is stored in the suffix string. Using .jpg on a 4-channel image will not work, as JPEGs can only store 3 colour channels.

Default file prefix:

DefaultMaterial_

Default file suffix for green:

Roughness.png

Batch Import/Export – for advanced users

Under the separated channels you'll notice batch import and export buttons. These open a folder selection screen. Once you select the folder, they'll import or export based on the file name prefix and suffix set by the user.

For example, if the default prefix is set to "WeaponTexture_" and the green suffix is set to "Roughness.png", then the batch importer will look for a file named "WeaponTexture_Roughness.png" to put in the green channel.

If the files are missing or named incorrectly, they won't import. If alpha is disabled, it won't look for the file and won't try to import a texture.

As of the current version (1.0.2), batch import is currently incompatible with the autopacker. This is a bug I plan on fixing soon.



Batch import channels



Batch export channels

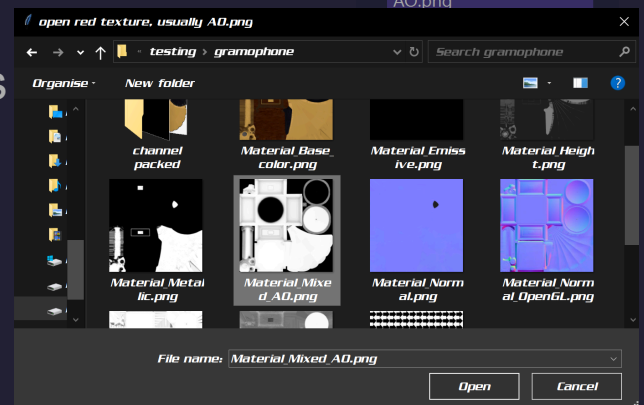
Common Use Cases

Packing RGB(A) into a single image

1. Import the unpacked textures using the buttons under the image channels. The open file window name will give you hints of which PBR texture is recommended for the channel.
2. If using the alpha channel, remember to toggle 'Use Alpha'.
3. If autopack is disabled, press Pack Textures.
4. Export the packed texture using the button under the packed image.

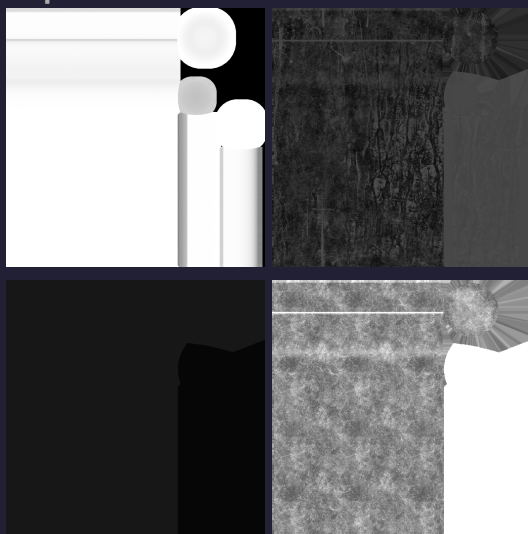


Import red texture
Export red texture
Default file suffix for red:
AO.png

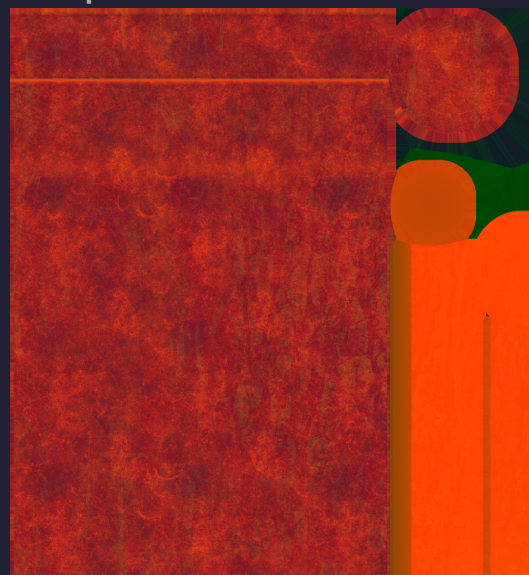


Results:

Inputs:



Output:



Unpacking RGB(A) from prepacked texture

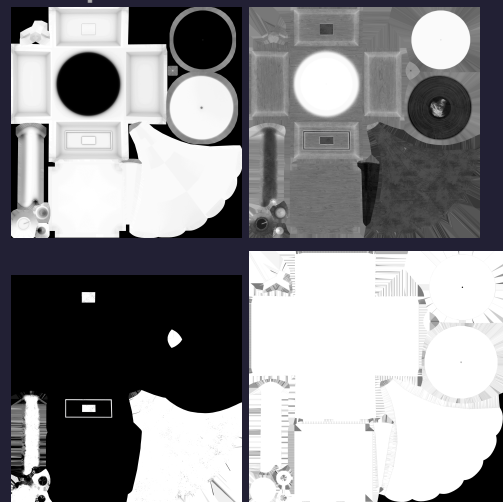
1. Open a packed image using the 'import packed texture' button under the packed image on the right.
2. If autopacker is disabled, click 'Unpack Textures'.
3. If you want to unpack the alpha, you may need to toggle the 'Use Alpha' tickbox and click 'Unpack Textures'.
4. Export the individual channels using the export button located under the image.
5. Alternatively, set the default file prefix and suffixes and use the batch export button to export them all at once

Results:

Input:



Outputs:



Adding a single channel to an existing image

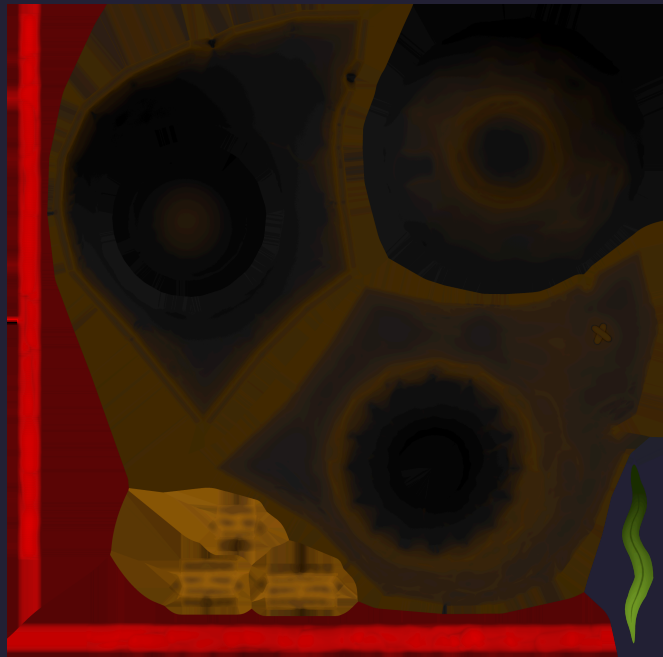
This is often used to add an alpha mask to an existing image

1. Open a texture in the packed texture slot and unpack if autopacker is disabled.
2. If you want to add the alpha texture, toggle 'Use Alpha' to true.
3. Replace the individual channel using the import button under the image and repack if necessary.
4. Export the packed texture

Inputs:



Output:

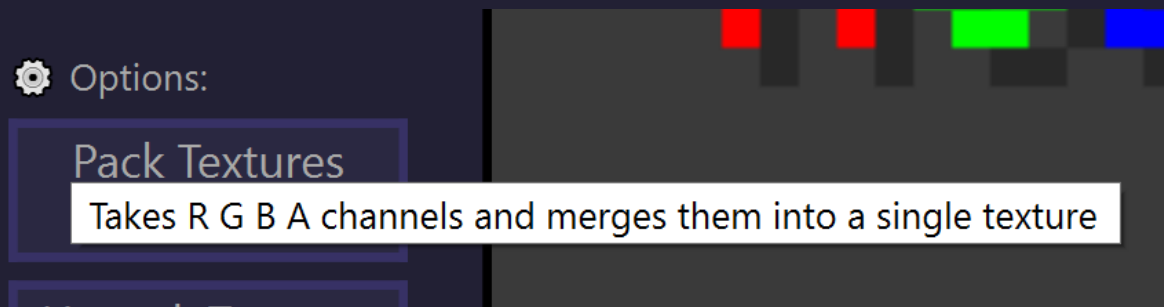


Troubleshooting

Tooltips

If you're unsure what a button does, hover your mouse over it for a tooltip.

An example is shown below



Default output

When there are any errors with exporting a texture to a selected location, the tool will automatically save to a default output folder. The file is saved in:

tool_location/Channel_Packing_Tool/default_output and will use the file prefix and suffix assigned by the user. This is a fallback location, so it's worth checking if you get any crashes etc.

Textures not packing

Textures with different resolutions will not pack together. If you're not using the alpha, then consider disabling it and see if it was failing to pack because the unused alpha was the wrong resolution.

Known issues – see bug tracker for more information

====Known issues:

GUI: Images scaling breaks aspect ratio
| Priority: Low
| UPDATE: Kinda works now but jittery and stretches when made smaller.
| changed priority to low as it's functional
| Aspect ratio forced 1:1 so different images will display as square. chosen to keep it this way for visual consistency
| still correct size when exporting. only affects GUI

Tool: Images of different resolutions can't pack together
| Priority: Medium

Tool: Alpha still unpacking when disabled
| Priority: Low

Tool: Gives an error message when attempting to export a .jpg file with 4 Channels
| Priority: Medium

GUI: Cursor isn't updated to hand pointer when hovering over a button
| Priority: Low

GUI: Save/open windows don't close when closing the main windows
| Priority: Low

Tool: Autopacker doesn't work on batch import
| Priority: Medium

GUI: Import and export windows don't use the main icon
| Priority: Low

Tool: Jpeg imports don't always keep rotation data. Not huge issue as you can just export and right click > rotateleft/right
| Priority: Low

Development Information

Roadmap

1.0.0: Initial release

1.0.1: Bug fixes
User guide

1.0.2: Bug fixes ← Current version

1.0.3: Bug fixes

1.1.0: Feature release
Warning popup for resolution
scaling
Overhaul name system to add
dropdown for file extension

1.1.1: Bug Fixes

```
# Version: 1.0.2
#           ^ ^ ^
# release  └─┘ ┘
# version  ┘ ┘
# (1 or 0) ┘ ┘
#           ┘ ┘
# feature  ┘ ┘
# version  ---└─┘
#           ┘ ┘
# bugfix   ┘ ┘
# version  -----└─┘
```

```
=====Features=====
==Planned Features:
|
| Tool: Error Warning Window
| Priority: High
|
| Tool: Resolution scaling
| Priority: High
|
| Tool: bit depth export
| Priority: Low
|
| Tool: Channel Invert Button
| Priority: Medium
|
| Tool: sepearate suffix from name (as dropdown for format type):
| Priority: High
|
=QOL:
|
| GUI: Langer, nicer Font
| Priority: Medium
|
| GUI: Warning when using absolute values in image. e.g. total black and total white
| Priority: High (following Feedback, moved from low)
|
=Easter Eggs:
|
| GUI: different icon for world events. e.g. pride/halloween/xmas
| Priority: Low
|
==WIP Features:
|
| Tool: Default file Names
| need to fix
```

For each of these
versions I'd assign
between 3 days and a
week

See bugtracker.txt for
more planned
features

Credit

Special Thanks:

Omar Zohdi – for teaching me Python and accommodating my learning differences where applicable.

Escape Studios user testers – for their invaluable feedback and in-person reactions.

Studio 694 (WCUC) – for more invaluable feedback

Leamington Game Dev Group – for even more invaluable feedback

AI Statement

While I actively avoided using any generative AI in my code for moral, ethical and environmental reasons, much of the PySide6 documentation was automatically translated from C, which may have used AI. Additionally, when documentation wasn't enough, I often checked Reddit or other internet forums for help. I can't verify if these sources used AI or not, but I actively avoided ones outright admitting to AI use.

I have done my best to avoid AI in all aspects of this project and can confidently claim this is my own work.

Libraries used:

Preinstalled:

- Sys
- Os – for paths
- Random – for splash text
- Functools – to pass variables through button inputs
- Tkinter – for built-in file open and save windows

Installed

- Pyside6 – for GUI

```
pip install pyside6
```

- Pillow – for image manipulation

```
pip install Pillow
```

- Pyinstaller – for packaging into an executable file

```
pip install pyinstaller
```